

1. Consider an undirected weighted network consisting of n vertices, where each ~~vertex~~^{edge} is assigned a weight α , and let A denote its adjacency matrix.

- (a) What is the total number of paths of length r between two vertices s and t ?
- (b) What is the sum of the weights of paths of length r between s and t ?
- (c) What is the sum of the weights of all paths of length r or less between s and t ?
- (d) What does this last sum converge to in the limit $r \rightarrow \infty$? (Hint: You did this in calculus, but with a scalar variable x rather than a matrix.)

[Marks: $1 + 1 + 1 + 2 = 5$]

2. Consider the Allee-effect population model

$$\frac{dN}{dt} = rN(-N^2 + aN - b), \quad (1)$$

where $r, a, b > 0$.

- (a) Find all fixed points of Eq. (1). Show that a nontrivial pair of fixed points exists only when $a^2 > 4b$, and give their values. Determine the stability of each nontrivial fixed point via linear stability analysis.
- (b) Now consider the model in Eq. (1) again. Show that the characteristic timescale for the largest- N stable fixed point is given by

$$\tau = \frac{1}{-r \left(-\frac{a^2}{2} + 2b - a \sqrt{\frac{a^2}{4} - b} \right)}.$$

[Marks: 4 + 1 = 5]

3. Consider a network whose graph Laplacian is given by

$$L = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

The diffusion dynamics on the network are governed by

$$\dot{\mathbf{x}} = -CL\mathbf{x}, \quad C > 0,$$

where $\mathbf{x}(t)$ represents the state of the four nodes.

- (a) Without explicitly calculating the determinant of L , identify an eigenvalue and its corresponding eigenvector.
- (b) Suppose the initial state can be expanded in terms of the Laplacian eigenmodes as

$$\mathbf{x}(0) = \sum_{r=1}^4 a_r(0) \mathbf{v}_r,$$

where

$$L\mathbf{v}_r = \lambda_r \mathbf{v}_r.$$

Show that the contribution of each eigenmode evolves as